MAHMOUD SALAMA

SOLUTION & ENTERPRISE ARCHITECT | SOFTWARE ENGINEERING LEADER

Solution & Enterprise Architecture | Software Engineering | APIs & Integration | Data Platforms

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PROFESSIONAL SUMMARY

Solution and enterprise architect with 18+ years across Egypt and Oman, progressing from hands-on Linux/web and full-stack engineering to development leadership and national-scale platform transformation. Personally developed C#/.NET applications and healthcare REST APIs; governs modern ReactJS/Node.js, PostgreSQL, Redis and containerized platform delivery through specialist teams. Leads a 54-person organization with 9 direct reports, directing applications, integration, data, DevOps and resilience.

CORE SKILLS

Enterprise & Solution Architecture | Target-State & Transition Architecture | Cloud-Native Modernization | Integration & API Architecture | Data Architecture & Governance | NFRs & Production Readiness | Security & Resilience by Design | Architecture Risk & Technical Debt | Build-vs-Buy & TCO | Architecture Governance

SELECTED ARCHITECTURE IMPACT

• Led the end-to-end re-architecture and replacement of MedIQ for Egypt's Unified Procurement Authority (UPA), creating a national healthcare procurement, supply and inventory ecosystem. The transformation won the Pharmaconex Awards 2026 – Excellence in Digital Transformation.

• **Own architecture governance for MedIQ, a national-scale platform serving 38,000+ users across 5,900+ government healthcare entities and nearly 90,000 affiliated operational locations.**

• Architected MedIQ E-Tender as a governed enterprise workflow spanning 10 major steps, 30 operational stages, 15 Chairman approvals and 20+ roles/permission groups, with RBAC, segregation of duties, auditability and sensitive approval controls.

• Directed the completed Microsoft Dynamics 365 enterprise integration across procurement, supply, inventory, finance and reporting, governing data ownership, API/interface contracts, mapping, reconciliation, exception handling, security, change control and production transition.

• Led UPA-side architecture for the greenfield, bi-directional SAP Strategic Medical Warehouses integration, defining system boundaries, integration approach, master-data harmonization, mapping, security and governance across request and financial workflows.

PROFESSIONAL EXPERIENCE

Egyptian Authority for Unified Procurement (UPA) | Cairo, Egypt 05/2025 - Present

Head of the Central Administration for Information Systems & Digital Transformation

Scope: *Technology Strategy | Enterprise Architecture | Digital Platforms | Integration & Data | Security/Resilience | Delivery Governance*

• Own technology strategy and enterprise architecture direction across applications, data, integrations, infrastructure and digital services, with EGP 150M annual IT budget accountability; define target states, architecture principles, investment priorities, transition sequencing and multi-year roadmaps.

• Lead a 54-person technology organization with 9 direct reports spanning software/applications, analysis, QA/reporting, DevOps, mobile, data, support/customer service and delivery governance.

• Led UPA-side architecture and delivery of the greenfield, bi-directional SAP Strategic Medical Warehouses integration, owning requirements, integration architecture, master-data harmonization/mapping, security, UAT, go-live and production support across request and financial workflows.

• Direct architecture and design decisions across engineering, QA, data, DevOps, infrastructure, security and vendors; use PoCs to assess architecture risk, technical debt and build-vs-buy and technology choices based on TCO, capability and strategic fit.

• Architected and governed MedIQ integrations with ECIC, MoH/HMIS, GAHAC and Blood Bank systems; ECIC and Blood Bank are confirmed production integrations. Also designed and developed production healthcare REST APIs for secure beneficiary, prescription and dispensing exchange with external health entities.

• Own resilience and production-readiness architecture including Private Cloud leadership, capacity planning, performance/load testing, PostgreSQL production/replica architecture, Redis caching, containerized services, F5 BIG-IP controls, DR and backup/off-site recovery controls.

• Translate legal, regulatory and governance requirements into architecture controls including least privilege, RBAC, segregation of duties, audit trails, traceability, retention, data protection and controlled approvals.

• Drive enterprise data and AI architecture across ODS and a production Central Data Warehouse; governed migration and reconciliation across invoice records and a controlled DeepSeek NL-to-SQL capability with read-only SELECT execution, SQL validation, allowlisted scope and RBAC. Designed and developed the HTA technology platform, covering solution architecture, system design and delivery.

Egyptian Authority for Unified Procurement (UPA) | Cairo, Egypt 10/2020 - 05/2025

General Manager - Systems, Applications & Technical Support

• Drove progressive modernization from established .NET Framework/Telerik applications toward ReactJS/Node.js services, C#/.NET APIs, PostgreSQL, Redis, containerized environments and Apache Superset while preserving operational continuity.

• Converted manual, paper- and spreadsheet-heavy processes into governed digital workflows with role-based access, multi-level approvals, validation, audit trails, reporting and clear operational ownership.

• Translate stakeholder needs into AS-IS/TO-BE workflows, functional requirements, acceptance criteria, API contracts and data mappings; apply UML/Visio and Jira across design, engineering, QA/UAT and production transition.

• Strengthened SDLC, release and production-readiness controls including architecture/design reviews, engineering standards, test gates, incident escalation, root-cause analysis, documentation and knowledge transfer.

Integral Solutions LLC | Muscat, Oman 05/2016 - 10/2020

Development Team Leader & Senior Full-Stack Developer

• Led a 20-person development team across 15+ Omani government and enterprise platforms while remaining hands-on in software engineering and technical delivery: requirements analysis, solution design, C#/.NET and ASP.NET development, SQL/database design, integrations, testing, deployment, troubleshooting and production support.

• Contributed extensively to Oman's National Educational Portal (~1M users), delivering LMS capabilities across courses/content, exams, attendance, student/teacher accounts, RBAC, reporting and integrations; also contributed to Maktabi and the Ministry of Education Recruitment System. Developed Maktabi correspondence workflows, reports and database components, with functional testing, release validation and production support.

• Designed secure government workflows, reporting, database and integration components across C#/.NET, ASP.NET Web Forms, SQL Server and MySQL.

Keyframe Egypt | Cairo, Egypt 04/2011 - 05/2016

Development Team Leader & Senior Full-Stack Developer

• Delivered or led 150+ major and mid-sized engagements across diverse sectors, progressing through hands-on full-stack engineering, application architecture and development-team leadership across multilingual, data-driven and integration-heavy applications.

• Owned analysis, C#/.NET engineering, database/reporting design, testing, deployment and support across multilingual and RTL solutions, including Yashfeen, a healthcare solution recognized through an ITIDA competition/award program. Selected historical projects include Rate Your Services and Kasr Al Ainy Liver Tumor Unit Management System.

Egyptian Armed Forces | Egypt 04/2008 - 10/2010

Linux Systems & Web Developer

• Built a systems-architecture foundation through Linux From Scratch, kernel modules, optimization/hardening and secure ASP.NET/SQL Server application development.

ARCHITECTURE CAPABILITIES & TECHNOLOGY STACK

Architecture & Transformation: Enterprise/Solution Architecture, business capability alignment, current/target state, architecture principles, transition roadmaps, application portfolio thinking, technology operating-model evolution, architecture risk and technical debt.

Integration & Data: API contracts, REST/SOAP/GraphQL, JSON/XML, mapping/transformation, data models, master/reference data, data ownership, quality and reconciliation.

Security, Resilience & Operations: Defense in depth, RBAC, least privilege, segregation of duties, TLS, network segmentation, resilience/DR, backup/recovery, capacity planning, performance testing and incident/RCA governance.

Technology Stack & Methods: C#/.NET, ASP.NET, ReactJS, Node.js, PostgreSQL, SQL Server, MySQL, Redis, Docker; Kubernetes (working knowledge), OpenAPI/Postman, Git/GitHub, Jenkins/TFS, UML/Visio, Jira, Agile/Waterfall and CI/CD. Architecture exposure: microservices/containerized-service patterns and cloud-native modernization.

EDUCATION

B.Sc. in Computer Science, Faculty of Computers & Informatics, Zagazig University | 2003-2007 | Arabic: Native | English: Professional Working Proficiency